5(a) State Lenz's law and explain how you would use a coil and a magnet to demonstrate the law. Make clear the use of any other apparatus in your explanation. [4]

(b) An instrument based on induced electromotive force (e.m.f.) is used to measure the speed of a projectile.

A small, strong but light magnet is imbedded in the projectile as shown in Fig. 5.1. The projectile passes through the two identical coils with *N* turns each, separated by a distance *d* with constant speed. As the projectile passes through each coil, a pulse of e.m.f. is induced in the coil. The time interval between the pulses can be measured accurately with an oscilloscope.

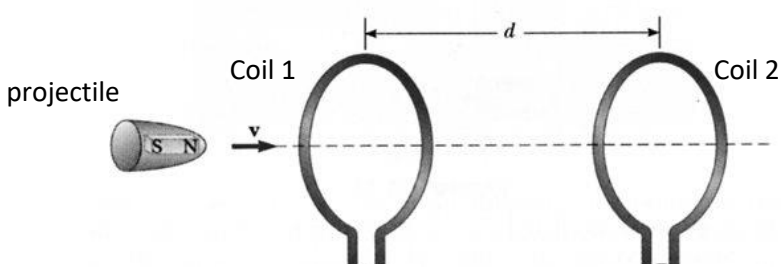

Fig 5.1

(i) Use the laws of electromagnetic induction to explain why an e.m.f. is induced in the coils. [2]

(ii) In Fig. 5.2, draw clearly a graph of the variation of induced e.m.f. against time for the arrangement shown in Fig. 5.1.

On your graph, label clearly which pulse is from coil 1 and which is from coil 2.

Fig. 5.2

(iii) If the pulse separation is 2.40 ms and *d* is 1.50 m, calculate the speed of the projectile. [1]

(iv) State and explain how the graph in Fig. 5.2 will differ if the speed of the projectile is increased. [2]